LIBRARY MANAGEMENT SYSTEM

Problem Statement

Managing a library manually can become difficult when the number of books and members increases. It can be challenging to keep track of available books, issued books, returned books, library members, and different book categories. Therefore, a simple computer-based system is required to perform basic library management operations in an organized manner.

Project Objective

The objective of this project is to develop a simple Library Management System using Python.

• View available books.

• Search for a book.

• Search for books according to their category.

• Issue a book to a library member.

• Return an issued book.

• View library members.

• View basic library statistics.

Proposed Solution

The proposed solution is a console-based Python application. The system will store book and member information using basic Python data structures such as dictionaries, tuples, and sets. Different functions will be used to perform library operations. The project will be divided into multiple Python files to keep the program organized.

Project Requirements

Books
The system should maintain information about at least 100 books. Each book should contain Book ID, book title, author name, and category.

Members
The system should maintain information about at least 50 library members. Each member should have a Member ID and member name.

Categories
The system should contain multiple book categories such as Programming, Computer Science, Networking, Artificial Intelligence, Machine Learning, Database, Cyber Security, Mathematics, Physics, Chemistry, Engineering, Electronics, Communication, and other relevant categories.

Functional Requirements

1. View Available Books
The system should display books that are currently available for issuing.

2. Search Book
The user should be able to search for a book using its title. The system should display the matching book and its current availability status.

3. Search by Category
The user should be able to enter a category and view the books belonging to that category.

4. Issue Book
The user should enter Book ID and Member ID. The system should check whether the book exists, the member exists, and the book is already issued. If all conditions are satisfied, the book should be issued.

5. Return Book
The user should enter the Book ID. The system should check whether the book is currently issued. If the book is issued, it should be marked as available again.

6. View Members
The system should display the registered library members.

7. Library Statistics
The system should display total books, available books, issued books, total members, and total categories.

Data Structures

Dictionary
Dictionaries will be used to store books and members.

```
books = {
    101: ("Python Basics", "Ravi Kumar", "Programming")
}
```

Tuple
Tuples will be used to store the fixed details of each book.

```
("Python Basics", "Ravi Kumar", "Programming")
```

Set
Sets will be used to store currently issued book IDs and available book categories.

```
issued_books = set()
```

File Handling

A text file named books.txt will be used to store basic information about book issue and return transactions. This provides basic experience with Python file handling.

```
101,5,Issued
101,Returned
```

Project Structure

```
Library-Management-System/
■
■■■ main.py
■■■ library.py
■■■ data.py
■■■ books.txt
■■■ README.md
■■■ statement.md
```

Module Description

main.py — Contains the main menu and controls the flow of the program.

library.py — Contains functions responsible for library operations.

data.py — Contains the books, members, issued-book information, and categories.

books.txt — Stores basic issue and return transaction information.

README.md — Contains documentation about the project.

statement.md — Contains the problem statement, objectives, requirements, and proposed solution.

Expected Outcome

After completing the project, the user should be able to perform basic library operations through a simple console interface. The project should demonstrate the practical use of Python dictionaries, tuples, sets, functions, loops, conditional statements, user input, exception handling, modules, and file handling.

Scope of the Project

This project focuses on basic library management operations. It is designed as a beginner-level Python project and does not require a graphical user interface, a database, external Python libraries, or internet connectivity. The project can be expanded in the future by adding advanced features such as database storage, login systems, due-date management, fine calculation, and a graphical user interface.

Conclusion

The Library Management System provides a simple way to demonstrate how basic Python programming concepts can be combined to solve a practical problem. The project focuses on writing understandable and organized code while providing the basic functionality required for managing books and library members.